Topic 12 – Governed, Optimized Workflow

Requirement Document

Project Name: Governed Enterprise Intelligence Assistant
Project Topic: Topic 12 – Governed, Optimized Workflow
Company Used: GreenTech Solutions Pvt. Ltd. (Fictional)
Document Type: Project Requirement Specification
Version: 1.0

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1. Project Overview

The Governed Enterprise Intelligence Assistant is designed to process enterprise data through multiple independent AI generation tasks and compile their outputs into a structured executive report.

The workflow incorporates monitoring, controlled authorization, and human-in-the-loop approval to support responsible enterprise reporting.

The system is intended to improve report generation efficiency while maintaining data accuracy, transparency, and approval governance.

2. Project Objectives

1. Accept enterprise data as workflow input.
2. Execute independent report-generation tasks in parallel.
3. Generate an enterprise report, data summary, and business insights.
4. Create monitoring and audit information for workflow execution.
5. Compile generated outputs into an executive review portal.
6. Collect an explicit manager approval decision.
7. Validate approval decisions before authorizing any controlled saving operation.
8. Restrict workflow sharing to authorized users.

3. Scope

3.1 In Scope

- Enterprise document input.
- Parallel AI generation tasks.
- Enterprise report generation.
- Data summarization.
- Business insight extraction.
- Monitoring and audit-log generation.
- Executive portal compilation.
- Manager approval input.
- Approval decision validation.
- Workflow sharing restrictions.
- Preview-based testing and verification.

3.2 Out of Scope or Pending Implementation

- Automatic saving of approved reports to shared company Google Drive.
- Automatic conditional routing after approval.
- Confirmed persistent centralized audit-log storage.
- Independently verified enterprise role-based access control.

These features remain outstanding in the current Opal implementation.

4. Input Requirements

Primary Input: `enterprise_raw_data`

The workflow must accept an enterprise data document containing relevant business information, such as:

- Financial performance.
- Revenue and expenses.
- Net profit and profit margin.
- Customer performance.
- Customer retention and satisfaction.
- Sales growth.
- Operational performance.
- Business challenges and opportunities.

The workflow should use the supplied data as its factual source and must not invent missing figures.

5. Functional Requirements

ID	Requirement	Description

FR-01	Enterprise Data Input	Accept an enterprise data document.
FR-02	Text Generation	Generate a structured enterprise report.
FR-03	Data Summarization	Extract and summarize important metrics and trends.
FR-04	Business Insight Extraction	Identify business insights, risks, and opportunities.
FR-05	Parallel Execution	Allow independent generation tasks to receive the enterprise input separately.
FR-06	Monitoring	Generate execution logs and workflow status information.
FR-07	Report Compilation	Combine available outputs into an executive review portal.
FR-08	Manager Approval	Request an explicit manager decision.
FR-09	Approval Validation	Validate APPROVE, REJECT, and invalid or missing decisions.
FR-10	Controlled Authorization	Mark saving as authorized only after a valid APPROVE decision.
FR-11	Governance	Never claim that an unconfigured saving operation has succeeded.

6. Non-Functional Requirements

6.1 Accuracy

- Preserve the original enterprise figures.
- Avoid unsupported claims and invented metrics.
- Distinguish source facts from generated recommendations.

6.2 Reliability

- Handle unavailable or incomplete input carefully.
- Identify execution errors where available.
- Avoid treating an empty model response as a successful result.

6.3 Security

- Restrict workflow sharing using available Opal sharing settings.
- Do not assume that sharing restrictions provide complete enterprise RBAC enforcement.
- Do not authorize saving without explicit approval.

6.4 Transparency

- Display the approval status clearly.
- Identify unavailable integrations.
- Do not claim successful storage without confirmation from a storage operation.

7. Output Requirements

The expected executive review portal should contain:

1. Executive Summary.
2. Financial Performance.
3. Customer Performance.
4. Operational Performance.
5. Business Insights.
6. Monitoring and Audit Log.
7. Human Approval Status.

The final output should be structured, readable, and suitable for executive review.

8. Approval Requirements

The Manager Approval input must accept an explicit decision.

Manager Input	Validation Status	Saving Authorization
APPROVE	APPROVED	AUTHORIZED
REJECT	REJECTED	BLOCKED
MAYBE or another invalid value	INVALID	BLOCKED
Blank or missing	INVALID	BLOCKED

The validation agent must never assume approval or convert a rejected or invalid decision into an approved decision.

9. Governance Model – 5:3:1

The project follows the 5:3:1 governance concept:

- **5 – Monitoring:** Monitor workflow execution and prepare audit information.
- **3 – Actuation:** Restrict write or save operations to controlled steps.
- **1 – Human-in-the-Loop:** Require manager review and approval before controlled action.

10. Acceptance Criteria

The project is considered to have passed its core functional tests when:

- Enterprise input is accepted.
- Independent generation tasks receive the source input.
- The executive portal contains the required sections.
- Extracted figures match the source data.
- Monitoring information is generated.
- APPROVE returns APPROVED and AUTHORIZED.
- REJECT returns REJECTED and BLOCKED.
- Invalid decisions return INVALID and BLOCKED.
- Blank approval returns INVALID and BLOCKED.
- The system does not falsely claim that a report has been saved.

11. Known Implementation Constraints

The current Opal environment did not expose a Google Drive save action or a conditional branching tool during implementation. Therefore, automatic report storage and approval-based routing are not considered completed requirements.

12. Conclusion

This requirement document defines the intended functionality and governance expectations of the Governed Enterprise Intelligence Assistant. The current implementation demonstrates the core report-generation, monitoring-output, and human approval validation capabilities, while identifying the remaining integration and routing requirements.